Kinetics of Front-End Acetylene Hydrogenation in Ethylene Production

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The kinetics of acetylene hydrogenation in the presence of a large excess of ethylene was studied in a laboratory flow reactor. Experiments were carried out using a $Pd/\alpha\text{-}Al_2O_3$ commercial catalyst and a simulated cracker gas mixture ($H_2/C_2H_2 = 50$; 60% C_2H_4; 30% H_2, and traces of CO), at varying temperature (293−393 K) and pressure (2−35 atm). Competing mechanisms for acetylene and ethylene hydrogenation were formulated and the corresponding kinetic equations derived by rate-determining step methods. A criterion based upon statistical analysis was used to discriminate between rival kinetic models. The selected equations are consistent with the adsorption of C_2H_2 and C_2H_4 in the same active sites followed by reaction with adsorbed hydrogen atoms to form C_2H_4 and C_2H_6 in a one-step process. Good agreement between computed and experimental results was obtained using a nonisothermal reactor model that takes into account the existence of external temperature and concentration gradients. The derived kinetic equations together with a pseudohomogeneous model of an integral adiabatic flow reactor were employed to simulate the conversion and the temperature profiles for a commercial hydrogenation unit.

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Introduction

The selective hydrogenation of acetylene in the presence of large amounts of ethylene is a process of considerable importance in the manufacture of polymer-grade ethylene. Small amounts of alkyne, in the parts per million range, affect the polymerization catalyst. Thus the acetylene concentration in the product stream must be reduced to 5 ppm or less.

This hydrogenation process could be located at different points in the purification section of an ethylene plant (Lam and Lloyd, 1972; Derrien, 1986). In one scheme the converter is placed after the conversion section, following a caustic scrubbing treatment to eliminate CO_2. Another alternative involves the hydrogenation of C_2H_2 in the C_2H_4-rich stream taken from the top of the de-ethanizer. The first alternative is known as front-end hydrogenation and the second one as tail-end hydrogenation.

Quite different operating conditions are used in each case. The front-end configuration involves the hydrogenation of C_2H_2 in the raw cracked-gas mixture so that the stream has a high H_2/C_2H_2 (≈100) ratio and several acetylenic, olefinic, and diolefinic components. Carbon monoxide also appears in the feed due to the inverse water-gas shift reaction in the cracking furnaces. They all act as inhibitors of C_2H_4 hydrogenation. Pure C_2H_2 in the tail-end converter is dosed with H_2 to obtain a stoichiometric H_2/C_2H_2 ratio.

Supported palladium catalysts with low metal content are used for both processes due to their exceptional high activity and selectivity. It is important to point out that Pd is unique, among other metals, for the preferential hydrogenation of alkynes and diolefins in the presence of olefins. However the $C_2H_4 \rightarrow C_2H_6$ reaction becomes significant when the C_2H_2 concentration is low, that is, at high levels of conversion.

Despite the commercial importance of C_2H_4 purification processes, there is limited kinetic information on the hydrogenation of C_2H_2 in the presence of C_2H_4. Most studies have been performed with pure C_2H_2 under conditions far removed from industrial operations. Some papers deal with the hydrogenation of $C_2H_2-C_2H_4$ mixtures (McGown et al., 1978; Moses et al., 1984; Adúriz et al., 1990). Orders in H_2 lie between 1 and 1.6, and that for C_2H_2 has been found to be zero or negative. Activation energy values between 9 and 16 kcal/mol have been quoted. For C_2H_4 hydrogenation on Pd the information is more scarce. Schuit and Van Reijen (1958) reported an activation energy of 8.4 kcal/mol, a H_2 order of 0.6, and an C_2H_4 order of 0, for Pd on SiO_2. Weiss et al. (1984) have investigated the effect of CO on C_2H_2 and C_2H_4 conversion. The former was found to be 50−60 times lower in the presence of CO (800 ppm) than in its absence, but the effect was small at low H_2 concentration. On the other hand, the hydrogenation of C_2H_4 was almost suppressed by the addition of CO.

The tail-end hydrogenation process, under conditions similar to those of industrial reactors, has been studied in a pilot plant by Battiston et al. (1982). The experimental results were correlated by an empirical model to take into account the influence of the several reaction variables on C_2H_2. Increasing the H_2/C_2H_2 ratio, at constant CO concentration, increases the conversion of both C_2H_2 and C_2H_4. A more standard approach was used by Men'shchikov et al. (1975). A commercial Pd catalyst was tested in a flow reactor at 20 atm, in the 353−430 K temperature range using a tail-end mixture. Rate equations for C_2H_2 and C_2H_4 hydrogenation, which were written under the assumption that the reactions occur on separate sites, gave good correlation of experimental data.

Studies of this kind for front-end C_2H_2 hydrogenation have not been reported. In this paper, laboratory data obtained on a wide range of pressure, temperature, and conversion values were fitted to several kinetic equations for C_2H_2 and C_2H_4 hydrogenation. The selected models were used to simulate the operation of an industrial acetylene converter. Predicted conversion and temperature profiles were compared with plant data.

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Methods

In the present study the experimental and theoretical work has been organized as follows:

1. Several reaction schemes were written for C_2H_2 and C_2H_4 hydrogenation on Pd using information from fundamental kinetic studies.
2. The rate-determining step method was applied to the proposed mechanisms in order to derive the corresponding kinetic equations.
3. Conversion versus temperature curves for C_2H_2 and C_2H_4 hydrogenation were obtained in a laboratory reactor, in the 1−30 atm pressure range, using a commercial Pd/α-Al_2O_3 catalyst.
4. The rate equations were coupled to three reactor models of increasing complexity. The algebraic and differential equations, which are not linear in the kinetic parameters, define a set of regression models.
5. A nonlinear regression routine that selects the conversion as a response variable was used to estimate the unknown parameters.
6. The discrimination between rival models was based upon the requirement of positive parameters supplemented with a variance analysis performed on the conversion residuals.

Kinetic Models

The first task of the present study was to derive kinetic equations that could be compared with experimental data for parameter estimation and discrimination. Several sequences of elementary steps were written on the basis of information available from previous studies. It is well-known that H_2 is dissociatively adsorbed on Pd to form H−Pd surface entities. On the other hand, it is accepted that C_2H_2 and C_2H_4 adsorption occurs by an associative mechanism involving one or two Pd surface atoms (π-bonded or σ-bonded structures). Consequently it was considered that similar or different types of sites may exist for H_2 and hydrocarbon adsorption. For the interaction of CO with the metal surface, an associative mechanism involving hydrogen adsorption sites has been postulated.

Regarding the surface reactions, three different reaction paths were assumed: a stepwise addition of adsorbed hydrogen atoms, the simultaneous addition of two atoms, or the reaction of gaseous hydrogen with adsorbed hydrocarbons. It has already been shown (Adúriz et al., 1990; Sarkany et al., 1986) that C_2H_6 formation during hydrogenation of C_2H_2−C_2H_4 mixtures is mainly due to C_2H_4 hydrogenation. Consequently we have excluded the direct formation of C_2H_6 from C_2H_2. Reaction schemes 1−5, shown in Table 1, were written on the basis of these considerations. In principle all steps were assumed to be reversible. If the stepwise addition of adsorbed hydrogen and the Eley−Rideal mechanism operate simultaneously (schemes 1 and 3, respectively), scheme 6 is obtained. On the other hand,, the combination of schemes 2 and 3 leads to scheme 7.

In order to obtain the steady-state kinetic models the rate-determining-step method was used. In principle any step can be rate limiting, but it is possible to reduce the number of equations on the basis of information available from surface chemistry studies. Hydrogen adsorption is a fast, nonactivated process, and therefore this step could be assumed to be at equilibrium. The desorption of ethane cannot be the rate-limiting step, for ethylene hydrogenation, because it is readily displaced from the surface by the strongly bound hydro-

Table 1. Reaction Mechanism for the Selective Hydrogenation of C_2H_2 and C_2H_4

scheme 1	scheme 2	scheme 3	scheme 4	scheme 5	scheme 6	scheme 7
(1) A + S → AS	(1) A + S → AS	(1) A + S → AS	(1) A + S → AS	(1) A + S → AS	(1) A + S → AS	(1) A + S → AS
(2) H_2 + 2S → 2HS	(2) H_2 + 2S → 2HS	(2) H_2(g) + AS → ES	(2) H_2 + 2Z → 2HZ	(2) H_2 + 2Z → 2HZ	(2) H_2 + 2S → 2HS	(2) H_2 + 2S → 2HS
(3) AS + HS → AHS + S	(3) AS + 2HS → ES + 2S	(3) E + S → ES	(3) AS + 2HZ → ES + 2Z	(3) AS + HZ → AHS + Z	(3) AS + HS → AHS + S	(3) AS + 2HS → ES + 2S
(4) AHS + HS → ES + S	(4) E + S → ES	(4) H_2(g) + ES → EtS	(4) E + S → ES	(4) AHS + HZ → ES + Z	(4) AHS + HS → ES + S	(4) AS + H_2(g) → ES
(5) E + S → ES	(5) ES + 2HS → EtS + 2S	(5) Et + S → EtS	(5) ES + 2HZ → EtS + 2Z	(5) E + S → ES	(5) AS + H_2(g) → ES	(5) E + S → ES
(6) ES + HS → EHS + S	(6) Et + S → EtS	(6) CO + S → COS	(6) Et + S → EtS	(6) ES + HZ → EHS + Z	(6) E + S → ES	(6) ES + 2HS → EtS + 2S
(7) EHS + HS → EtS + S	(7) CO + S → COS		(7) CO + S → COS	(7) EHS + HZ → EtS + Z	(7) ES + HS → EHS + S	(7) ES + H_2(g) → EtS
(8) Et + S → EtS				(8) Et + S → EtS	(8) EHS + HS → EtS + S	(8) Et + S → EtS
(9) CO + S → COS				(9) CO + S → COS	(9) ES + H_2(g) → EtS	(9) CO + S → COS
					(10) Et + S → EtS	
					(11) CO + S → COS	

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Table 2. Kinetic Equations Derived from the Reaction Schemes Presented in Table 1[a]

$r_{2,14} = r_{7,15}$	$r_A = k_A C_A/D$ $r_E = k_E C_E/D$ $D = 1 + (K_H C_H)^{1/2} + C_{Et} \sum_{i=1}^{2} K_i CH^{-i} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{1,36}$	$r_A = k_A C_A C_H^{1/2}/D^2$ $r_E = k_E C_E C_H^{1/2}/D^2$ $D = 1 + K_A C_A + K_E C_E + (K_H C_H)^{1/2} + (K_1 C_E + K_2 C_{Et}) C_H^{-1/2} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{1,47}$	$r_A = k_A C_A C_H/D^2$ $r_E = k_E C_E C_H/D^2$ $D = 1 + K_A C_A + K_E C_E + (K_H + K_1 C_A + K_2 C_E) C_H^{1/2} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{2,35}$	$r_A = k_A C_A C_H/D^3$ $r_E = k_E C_E C_H/D^3$ $D = 1 + K_A C_A + K_E C_E + (K_H C_H)^{1/2} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{3,13} = r_{4,14}$	$r_A = k_A C_A/D$ $r_E = k_E C_E/D$ $D = 1 + (K_{Et} + K_1 C_H^{-1} + K_2 C_H^{-2}) C_{Et} + K_{CO} C_{CO}$
$r_{3,24}$	$r_A = k_A C_A C_H/D$ $r_E = k_E C_E C_H/D$ $D = 1 + K_A C_A + K_E C_E + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{4,35}$	$r_A = k_A C_A C_H/D$ $r_E = k_E C_E C_H/D$ $D = (1 + K_A C_A + K_E C_E + K_{Et} C_{Et} + K_{CO} C_{CO})(1 + K_H^{1/2} C_H^{1/2})^2$
$r_{5,15}$	$r_A = k_A C_A/D$ $r_E = k_E C_E/D$ $D = 1 + (K_{Et} + K_1 C_H^{-1/2} + K_2 C_H + K_3 C_H^{-3/2} + K_4 C_H^{-2}) C_{Et} + K_{CO} C_{CO}$
$r_{5,36}$	$r_A = k_A C_A C_H^{1/2}/D$ $r_E = k_E C_E C_H^{1/2}/D$ $D = (1 + K_A C_A + K_E C_E + (K_1 C_E + K_2 C_{Et}) C_H^{-1/2} + K_{Et} C_{Et} + K_{CO} C_{CO})(1 + K_H^{1/2} C_H^{1/2})$
$r_{5,47}$	$r_A = k_A C_A C_H/D$ $r_E = k_E C_E C_H/D$ $D = (1 + K_A C_A + K_E C_E + (K_1 C_A + K_2 C_E) C_H^{1/2} + K_{Et} C_{Et} + K_{CO} C_{CO})(1 + K_H^{1/2} C_H^{1/2})$
$r_{6,16} = r_{1,15}$	$r_A = k_A C_A/D$ $r_E = k_E C_E/D$ $D = 1 + (K_H C_H)^{1/2} + (K_1 C_H^{-1/2} + K_2 C_H^{-1} + K_3 C_H^{-3/2} + K_4 C_H^{-2}) C_{Et} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{6,35/79}$[b]	$r_A = k_A C_A C_H^{1/2}/D^2 + k_A^* C_A C_H/D$ $r_E = k_E C_E C_H^{1/2}/D^2 + k_E^* C_E C_H/D$ $D = 1 + K_A C_A + (K_H C_H)^{1/2} + K_E C_E + (K_1 C_E + K_2 C_{Et}) C_H^{-1/2} + K_{Et} C_{Et} + K_{CO} C_{CO}$
$r_{6,45/89}$[b]	$r_A = k_A C_A C_H/D^2 + k_A^* C_A C_H/D$ $r_E = k_E C_E C_H/D^2 + k_E^* C_E C_H/D$ $D = 1 + K_A C_A + K_E C_E + (K_H^{1/2} + K_1 C_A + K_2 C_E)\, C_H^{-1/2} + K_{ET} C_{ET} + K_{CO} C_{CO}$
$r_{7,34/67}$[b]	$r_A = k_A C_A C_H/D^3 + k_A^* C_A C_H/D$ $r_E = k_E C_E C_H/D^3 + k_E^* C_E C_H/D$ $D = 1 + K_A C_A + (K_H C_H)^{1/2} + K_E C_E + K_{Et} C_{Et} + K_{CO} C_{CO}$

[a] The first digit identifies the sequence of steps and the others the controlling steps for C_2H_2 and C_2H_4 hydrogenation. [b] Simultaneously mechanisms with two different controlling steps for acetylene and ethylene hydrogenation are considered.

carbons (ethylene and acetylene). In addition it has been shown recently (Park and Price, 1991) that CO enhances the desorption of C_2H_4. Consequently only the adsorption of C_2H_2 and C_2H_4 and the surface interactions were chosen as rate-controlling steps to derive the kinetic models. To further reduce the number of equations, the same determining step was selected for C_2H_2 and C_2H_4 hydrogenation. By use of this approach 14 kinetic models were derived, as shown in Table 2.

As an example let us consider scheme 2 when the adsorption of C_2H_2 and C_2H_4, steps 1 and 4, are rate-controlling. The corresponding kinetic equations are

$$r_{2,1} = -r_A = k_1 C_A[S]$$

$$r_{2,4} = -r_E = k_4 C_E[S] \qquad (1)$$

where k_1 and k_4 are the rate constants and [S] is the concentration of free surface sites. Considering the irreversible nature of the overall hydrogenation reactions, the reverse reactions of steps 1 and 4 have been neglected. The remaining steps, 2, 3, 5, 6, and 7 are considered to be in a quasiequilibrium state. We use letters to identify the adsorption−desorption processes at equilibrium: K_H for H_2, K_{Et} for C_2H_6 , K_E for C_2H_4, and K_{CO} for CO, while the constants K_{AS} and K_{ES} take into account the surface reactions of adsorbed C_2H_2 and C_2H_4 at equilibrium. In order to simplify the rate equations the following constants were introduced

$$K_1 = \frac{K_{Et}}{K_H K_{ES}} \qquad K_2 = \frac{K_{Et}}{K_H^{\,2} K_{ES} K_{AS}} \qquad (2)$$

Using these constants, [S] may be written in terms of the gas phase concentrations

$$[S] = [S_T]/D$$

where

$$D = 1 + (K_H C_H)^{1/2} + C_{ET} \sum_{i=1}^{2} K_i C_H^{-i} + K_{ET} C_{ET} + K_{CO} C_{CO} \qquad (3)$$

and $[S_t]$ is the concentration of free and occupied sites.

Combining the previous equations, the rates of disappearance of C_2H_2 and C_2H_4 are

$$-r_A = k_A C_A/D \quad \text{and} \quad -r_E = k_E C_E/D \qquad (4)$$

where

$$k_A = k_1[S_t] \quad \text{and} \quad k_E = k_4[S_t]$$

Experimental Section

A laboratory scale reactor was used to obtain conversion vs temperature curves in the 1−30 atm pressure range. Catalyst charges of 1−3 g were packed in a

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Table 3. Feed Mixture and Operating Conditions for the Laboratory Reactor

components	composition (mol % ± 0.02)
CH_4	6.53
C_2H_2	0.52
C_2H_4	43.99
C_2H_6	26.72
H_2	22.1
CO	150−5000 ppm
flow rate	100 cm^3 (STP)/min
mass of catalyst	1−3 g
pressure	1−35 atm
temperature	288−393 K

Table 4. Catalyst Characterization

support	α-Al_2O_3
palladium content (wt %)	0.04
BET surface area (m^2/g)	21
metal dispersion (H/Pd)	0.15
apparent density (g/cm^3)	1.61
real density (g/cm^3)	3.61
average pore diameter (Å)	1000
pore volume (cm^3/g)	0.338
porosity (%)	54.5

stainless-steel tube (10 mm i.d.) immersed in a diethylene glycol bath. The total pressure was adjusted by means of a back-pressure regulator. Experiments were performed in the 288−393 K temperature range while the gas flow rate was held constant at 100 cm^3 STP min^{-1} using a mass flow controller. A thermocouple in contact with the catalyst particles was located in an axial position to monitor the reaction temperature. In addition the wall temperature was measured.

The analysis of reactants and products was performed by on-line gas chromatography using a Carbosieve S-II column (1/8 in. × 80 cm) held at 393 K. Fresh catalyst samples were pretreated in N_2 at 433 K for 8 h before feeding the hydrocarbon mixture.

The reaction mixture was prepared by mixing CP-grade hydrocarbons with high-purity hydrogen and CO. Table 3 summarizes the gas composition and the main operating conditions.

A commercial catalyst, ICI 38−1, containing 0.04% Pd, was used in this study. The main catalyst properties are shown in Table 4.

The effect of increasing temperature on C_2H_2 and C_2H_4 conversion was investigated in the presence of varying amounts of CO and a total pressure of 1, 9, 16, and 32 atm. In these runs the space velocity was held constant. When the conversion of C_2H_2 was almost 100%, the heat of reaction produced a small temperature difference between the catalyst bed and the reactor wall. The maximum C_2H_4 conversion was about 10−15%. A large experimental error was present in this measurement as compared with C_2H_2 conversion, due to the high concentration of C_2H_4 and C_2H_6 in the feed mixture. At high C_2H_4 conversion isothermal conditions could not be maintained and a large temperature gradient was detected by the thermocouples. In some experiments the temperature was held constant and the conversion was measured as a function of the total pressure.

The effect of temperature on C_2H_2 and C_2H_4 conversion, at 1 and 9 atm and 800 ppm CO, is shown in Figure 1. The initial trend is a large increase in C_2H_2 conversion within a narrow temperature range, followed by a marked decrease in the rate of reaction above 90% conversion. In this case the conversion of ethylene was observed after the complete elimination of C_2H_2, with a temperature span of about 15 K from $\chi_A = 100\%$ to $\chi_E = 1\%$. Consequently the Pd/α-Al_2O_3 catalyst, under the present experimental conditions, is very selective in the sense that it allows the total elimination of C_2H_2 with minimum C_2H_4 losses. Increasing the total pressure accelerates both reactions, and therefore similar conversions are obtained at lower temperatures. However a more detailed testing, in the region of high C_2H_2 conversion, indicates that the effect of pressure is more marked on the rate of C_2H_4 hydrogenation. Table 5 shows that the conversion of C_2H_2, at constant temperature, increases with pressure in the 1−9 atm range but remains constant afterward. On the other hand, the hydrogenation of C_2H_4 reflects a steady increase in conversion from 1 to 32 atm. Consequently the temperature span between complete elimination of C_2H_2 and the onset of C_2H_4 hydrogenation is reduced and the catalyst becomes less selective.

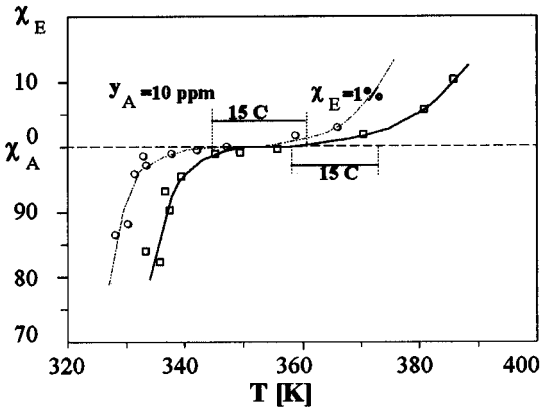


Figure 1. Acetylene and ethylene conversion as a function of temperature. Feed mixture as given in Table 3. SV = 33 cm^3 g_{cat}^{-1} min^{-1}. C_{CO} = 800 ppm. Pressure: □, 1 atm; ○, 9 atm.

Table 5. Effect of Pressure on C_2H_2 and C_2H_4 Conversion (CO Content = 1400 ppm; SV = 100 cm^3 STP g_{cat}^{-1} min^{-1})

		conversion (%)			
temp (K)		1 atm	8 atm	16.50 atm	32 atm
348	C_2H_2	85.54	95.86	95.39	94.13
	C_2H_4	0.00	0.00	0.00	0.00
359	C_2H_2	94.44	97.75	97.75	96.64
	C_2H_4	0.00	1.02	1.67	4.50
368	C_2H_2	96.13	98.16	98.02	96.97
	C_2H_4	0.00	3.12	6.07	14.50

Table 6. Effect of CO on C_2H_2 and C_2H_4 Conversion at Constant Pressure (Pressure = 16 atm; SV = 100 cm^3 STP g_{cat}^{-1} min^{-1})

		conversion (%)	
CO content (ppm)	temp (K)	C_2H_2	C_2H_4
150	341	98.31	10.45
1400	359	98.99	6.70
5000	393	98.62	6.08

The effect of CO on activity and selectivity, for experiments performed at 16 atm, is shown in Table 6. Increasing the concentration of CO, from 150 to 1400 ppm, lowered the C_2H_2 hydrogenation rate, and a higher temperature was needed to obtain nearly the same conversion. Despite the large increase in temperature, 18 K, the conversion of C_2H_4 decreased from 10.85% to 6.70%. When the CO concentration was raised to 5000 ppm, a further increase in temperature was required to maintain the same conversion of C_2H_2. However the conversion of C_2H_4 was almost the same: 6.08%. These results indicate that the concentration of CO at >150 ppm affects the rate of C_2H_2 and C_2H_4 hydrogenation to the same extent.

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Reactor Models

In order to process the experimental data, a suitable reactor model must be used. As a first approximation, a perfectly mixed flow reactor was considered (model I). Although this model is not appropriate to describe our laboratory setup, it has been chosen in an attempt to evaluate a large number of kinetic equations without considerable computing effort. The basic mass balance equations were written as

$$\frac{w}{F_i} = \frac{\chi_i}{r_i} \tag{5}$$

where i refers to C_2H_2 and C_2H_4. They can be readily coupled to the rate models given in Table 2 to relate the fractional conversion χ_i to the kinetic parameters. Therefore a set of nonlinear algebraic equations was obtained. Using experimental w/F_i and T values as independent variables and assuming tentative kinetic constants, the conversion ($\langle\chi_i\rangle$) was estimated and compared with the measured values (χ_i).

The second reactor model (model II) was assumed to be nonisothermal, isobaric with a plug flow pattern, and pseudohomogeneous. In this case the mass and energy balance defined a set of nonlinear ordinary differential equations coupled through the rate model r_i:

$$\frac{\mathrm{d}(vC_i)}{\mathrm{d}z} = (-r_i)\rho_{\mathrm{cat}} \tag{6}$$

$$c_p \frac{\mathrm{d}T}{\mathrm{d}z} = \rho_{\mathrm{cat}} \sum_i (-\Delta H_i)(-r_i) - 4U(T - T_{\mathrm{w}}) \tag{7}$$

The overall heat transfer coefficient U was defined as follows:

$$U = \left(\frac{1}{hA} + \frac{\ln(d_{\mathrm{to}}/d_{\mathrm{t}})}{2\rho k_{\mathrm{w}} A}\right)^{-1} \tag{8}$$

In this equation the heat transfer coefficient h takes into account the total resistance to heat flow in the radial direction and it is given by the Crider and Foss equations (Crider and Foss, 1965).

$$1/h = 1/h_{\mathrm{w}} + d_{\mathrm{t}}/6k_{\mathrm{ref}} \tag{9}$$

$$h_{\mathrm{w}} = 3k_{\mathrm{ref}} Re^{-0.25}/d_{\mathrm{t}} \tag{10}$$

where h_{w} is the heat transfer coefficient at the tube wall. The radial effective thermal conductivity k_{ref} was obtained from the Dixon and Cresswell correlation (Dixon and Cresswell, 1979)

$$k_{\mathrm{ref}} = k_{\mathrm{rf}} + k_{\mathrm{rs}}\left[\frac{1 + (8k_{\mathrm{rf}}/h_{\mathrm{wf}}d_{\mathrm{t}})}{1 + \dfrac{(16/3k_{\mathrm{rs}})\left(\dfrac{1}{h_{\mathrm{fs}}d_{\mathrm{p}}} + \dfrac{0.1}{k_{\mathrm{p}}}\right)}{(1-\epsilon)(d_{\mathrm{t}}/d_{\mathrm{p}})^2}}\right] \tag{11}$$

where k_{rf} and k_{rs} are the radial conductivities of the fluid and the solid, respectively. The effective conductivity depends also on the fluid/wall (h_{wf}) and the fluid/solid (h_{fs}) heat transfer coefficients. Suitable correlations for these coefficients are found in the previous reference.

The previous equations constitute an initial value problem that has been solved by means of a Runge–Kutta method. In this way, the C_2H_2 and C_2H_4 conversions ($\langle\chi_i\rangle$) at the reactor outlet are obtained.

A more complex reactor model (model III) emerges if one assumes the existence of temperature and concentration gradients in the gas phase region adjacent to the catalyst particles. Therefore $T \neq T_{\mathrm{s}}$ and $C_i \neq C_{si}$. This situation is likely to occur in the laboratory reactor due to the low gas velocity over the catalyst particles.

Under steady-state conditions the rate of mass transport of a reactant (i) through the gas phase film is equal to the observable surface reaction rate ($r_{\mathrm{s}i}$):

$$r_{si} = k_{\mathrm{m}i} a_{\mathrm{m}} (C_i - C_{si}) \tag{12}$$

Introducing the external effectiveness factor η_i,

$$r_{si} = \eta_i r_{\mathrm{io}} \tag{13}$$

where r_{io} is the maximum rate or reaction obtained when $C_{\mathrm{s}i} = C_i$. In addition the Damkohler number D_{io} is defined as the ratio of the maximum surface rate to the maximum mass transport rate:

$$D_{\mathrm{io}} = r_{\mathrm{io}}/(k_{\mathrm{m}i} a_{\mathrm{m}} C_i) \tag{14}$$

Following the treatment of Carberry (Carberry, 1976), eqs 13 and 14 can be related to define a dimensionless observable quantity:

$$\eta_i D_{\mathrm{io}} = r_{si}/(k_{\mathrm{m}i} a_{\mathrm{m}} C_i) \tag{15}$$

Combining eqs 12 and 15, a dimensionless concentration in terms of an observable quantity may be obtained:

$$C_i^* = C_{si}/C_i = 1 - \eta_i D_{\mathrm{io}}$$

For C_2H_2,

$$C_{\mathrm{A}}^* = C_{\mathrm{s}A}/C_{\mathrm{A}} = 1 - \eta_{\mathrm{A}} D_{\mathrm{Ao}} \tag{16}$$

Taking into account that ethylene is a product and also a reactant, the concentration on the gas–solid interphase is given by

$$C_{\mathrm{E}}^* = 1 - \eta_{\mathrm{E}} D_{\mathrm{Eo}} + \eta_{\mathrm{A}} D_{\mathrm{Ao}} k_{\mathrm{m}A} C_{\mathrm{A}}/(k_{\mathrm{m}E} C_{\mathrm{E}}) \tag{17}$$

On the other hand, the H_2 concentration is related to the consumption of both C_2H_2 and C_2H_4:

$$C_{\mathrm{H}}^* = 1 - \eta_{\mathrm{A}} D_{\mathrm{Ao}} k_{\mathrm{m}A} C_{\mathrm{A}}/(k_{\mathrm{m}H} C_{\mathrm{H}}) - \eta_{\mathrm{E}} D_{\mathrm{Eo}} k_{\mathrm{m}E} C_{\mathrm{E}}/(k_{\mathrm{m}H} C_{\mathrm{H}}) \tag{18}$$

Under steady-state, nonisothermal conditions, the surface temperature could be related to the bulk gas phase temperature through an energy balance on the catalyst particles.

$$h_{\mathrm{fs}} a_{\mathrm{m}} (T_{\mathrm{s}} - T) = (-\Delta H_{\mathrm{A}})(-r_{\mathrm{A}}) + (-\Delta H_{\mathrm{E}})(-r_{\mathrm{E}}) \tag{19}$$

Dividing this equation by ($k_{\mathrm{m}H} a_{\mathrm{m}} T$) and recalling the $j_{\mathrm{D}} = j_{\mathrm{H}}$ heat and mass transfer analogy,

$$\rho c_p (Sc/Pr)^{2/3}(T^* - 1) = (-\Delta H_{\mathrm{A}})(-r_{\mathrm{A}})/(k_{\mathrm{m}H} a_{\mathrm{m}} T) + (-\Delta H_{\mathrm{E}})(-r_{\mathrm{E}})/(k_{\mathrm{m}H} a_{\mathrm{m}} T)$$

$$T^* = 1 + [(-\Delta H_{\mathrm{A}})(-r_{\mathrm{A}})/(k_{\mathrm{m}H} a_{\mathrm{m}} T \rho c_p)](Pr/Sc)^{2/3} + [(-\Delta H_{\mathrm{E}})(-r_{\mathrm{E}})/(k_{\mathrm{m}H} a_{\mathrm{m}} T \rho c_p)](Pr/Sc)^{2/3} \tag{20}$$

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Using eq 15 the dimensionless temperature may be written

$$T^* = 1 + \beta_A \eta_A D_{Ao} + \beta_E \eta_E D_{Eo} \tag{21}$$

where

$$\beta_i = [(-\Delta H_i)(k_{mi} C_i)/(k_{mH} T \rho c_p)](Pr/Sc)^{2/3} \qquad i = \text{A, E}$$

The temperature dependence of the rate constant k_i and the adsorption equilibrium constant K_i may also be written in a dimensionless form:

$$k_i^* = k_{si}/k_i \text{ (or } K_i^* = K_{si}/K_i) = \exp(-E_i^*(1/T^* - 1)) \tag{22}$$

where:

$$E_i^* = E_i/RT$$

To integrate eqs 6 and 7 for a given axial coordinate, eqs 16−22 must be solved to find the concentrations and temperature at the catalyst surface. Consequently, a system of coupled nonlinear differential and algebraic equations has to be solved. In this work, the ordinary differential equations were integrated with the Runge−Kutta−Gill method and the algebraic equations were solved using a quasi-Newton algorithm, where the first Jacobian was numerically evaluated and then updated by means of Broyden's method (Paloschi and Perkins, 1988) at each iteration step. Thus the overall computing time was greatly reduced.

A nonlinear regression routine based on the Marquardt algorithm was used for the three reactor models described above to adjust the kinetic parameters corresponding to the rate models given in Table 2. The objective function to be minimized was defined as

$$\phi = \sum_i [\chi_i - \langle \chi_i \rangle]^2 \tag{23}$$

In the models discussed above we have neglected the presence of internal diffusion limitations. This simplifying assumption was based on the fact that the average pore diameter of the catalyst pellets was quite large, 1000 Å. However, due to the high hydrogenation rate and the homogeneous distribution of Pd in the pellet, internal concentration gradients may be present. In order to check for the absence of intrapellet diffusion limitations, we selected the criterion of Weisz and Prater (Froment and Bischoff, 1990) which in turn requires a measured value of the rate of reaction. Consequently application of this criterion was performed after adequate kinetic equations were available to calculate the rates of C_2H_2 and C_2H_4 hydrogenation at different conversion levels.

Results and Discussion

Determination of Kinetic Constants and Model Discrimination. The process parameters were considered as independent or input variables to be introduced as data in the mathematical models. On the other hand, the measured conversion values were the dependent variables to be compared with the theoretical predictions. From this comparison the regression routine estimates the kinetic parameters. The rate equations of Table 2 were first included in model I. The calculation starts with an initial guess for each parameter, and it continues until the variance analysis indicates that a reasonable agreement between experimental and calculated conversions has been found. A set of 163 experimental points was used for these estimations.

For the more complex reactor models, the procedure described above uses a large computing time due to the fact that the calculated conversion is obtained by numerical integration of differential equations. In addition each step of the regression routine requires about 2600−3200 simulations to evaluate the Jacobian. In order to reduce computing time, a shortcut method was adopted, based on sequential discrimination, so that many rate models were eliminated early in the process using model I, where only one algebraic equation needs to be solved. A set of parameters was estimated for each rate equation. The variance of the experimental and calculated conversion values provided a convenient method to analyze the quality of this fitting procedure. In addition the discrimination between the different rate equations was done on the basis of negative parameters or unexpected trends in the conversion versus temperature curves. In this way 11 rate models were eliminated, as shown in Table 7.

In the following stage the remaining kinetic equations were processed with the same regression routine but using model II. The parameters derived in the previous stage were now used as initial values. One kinetic model was eliminated when the discrimination criterion was applied. Finally the parameters determined with model II were adopted as initial values to fit the experimental data with model III. Therefore the adopted procedure uses a simple reactor model when the solution is a long way off, and the more realistic but time-consuming models were applied when few iterations were needed.

The Kinetic Equations. The best fit with the experimental data was provided by the rate model $r_{2,3}$ that assumes the adsorption of C_2H_2, C_2H_4, and H_2 on the same type of sites and the simultaneous addition of two hydrogen atoms to the absorbed hydrocarbons as determining steps. Table 8 presents the selected kinetic expressions and the parameter values obtained by minimization of the sum of residual squares. Upon examination these equations reveal some interesting features. First it is observed that the hydrocarbon adsorption constants K_E, K_{ET}, and K_A have a negligible effect on the rates and consequently the kinetic equations can be simplified to

$$-r_A = \frac{k_A C_A C_H}{[1 + (K_H C_H)^{1/2} + K_{CO} C_{CO}]^3} \tag{24}$$

$$-r_E = \frac{k_E C_E C_H}{[1 + (K_H C_H)^{1/2} + K_{CO} C_{CO}]^3} \tag{25}$$

It is observed that the rate of C_2H_4 hydrogenation does not depend on the concentration of C_2H_2 as previously reported by Margitfalvy et al. (1981). The explanation may be found in the presence of CO. As pointed out by Bos and Westerterp (1993a), CO may suppress the influence of C_2H_2 in C_2H_4 hydrogenation.

Another surprising feature is the positive values for the adsorption energy on the CO and H_2 constants which indicates that they will increase with temperature. It is reasonable to expect that the adsorption terms decrease with temperature due to the exothermic character of the adsorption processes. This anomalous

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Table 7. Discrimination Sequence for the Kinetic Models of Table 2

	model I		model II		model III	
r_{ii}	selection procedure	variance	selection procedure	variance	selection procedure	variance
$r_{2,14} = r_{7,15}$	*a*	0.366E−1				
$r_{1,36}$	*b*	0.481E−2	*c*	0.307E−2		
$r_{1,47}$	*a*	0.402E−2				
$r_{2,35}$	*b*	0.487E−2	*b*	0.270E−2	*b*	0.229E−2
$r_{3,13} = r_{4,14}$	*c*	0.319				
$r_{3,24}$	*a*	0.127E−1				
$r_{4,35}$	*a*	0.426E−1				
$r_{5,15}$	*c*	0.838E−2				
$r_{5,36}$	*a*	0.662E−2				
$r_{5,47}$	*a*	0.211E−1				
$r_{6,16} = r_{1,15}$	*a*	0.592E−2				
$r_{6,35/79}$	*a*	0.779E−2				
$r_{6,45/89}$	*a*	0.340E−2				
$r_{7,34/67}$	*b*	0.273E−2	*b*	0.239E−2	*c*	0.819E−2

a Adjusted by regression. [b] Rejected because $E_{act} < 0$.[b] [c] Rejected because of high variance.

Table 8. Selected Rate Equation and Kinetic Parameters for C_2H_2 and C_2H_4 Hydrogenation

kinetic parameters	log A_i	E (kcal/gmol)
k_A (m^6/(s g_{cat} kmol))	31.1	45.51
k_E (m^6/(s g_{cat} kmol))	26.6	42.94
K_H (m^3/kmol)	20.2	21.22
K_{CO} (m^3/kmol)	13.6	9.95
K_E (m^3/kmol)	0.26	0.005
K_{Et} (m^3/kmol)	−0.012	0.001
K_A (m^3/kmol)	−16	0.001

$$-r_A = \frac{k_A C_A C_H}{[1 + (K_H C_H)^{1/2} + K_A C_A + K_E C_E + K_{Et} C_{Et} + K_{CO} C_{CO}]^3}$$

$$-r_E = \frac{k_E C_E C_H}{[1 + (K_H C_H)^{1/2} + K_A C_A + K_E C_E + K_{Et} C_{Et} + K_{CO} C_{CO}]^3}$$

where $k_A = k_3 S_t^3 K_A K_H$ and $k_E = k_5 S_t^3 K_A K_H$

result may be due to data error, inadequate selection of experimental conditions, or data analysis. Problems of this kind are often encountered on rate modeling as discussed by Doraiswamy and Sharma (1984). In spite of this difficulty the hydrogenation rates predicted by our models exhibit a normal Arrhenius behavior because the effect of temperature on the adsorption parameters is overcompensated by the large activation energy on the rate constants k_A and k_E.

Equations 24 and 25 and those of model III were used in the simulation of an adiabatic reactor to observe the predicted rate versus temperature profiles, as shown in Figure 2. The rate of C_2H_2 hydrogenation is initially larger than that of C_2H_4, as expected. As the reaction proceeds, the decrease in C_2H_2 concentration is compensated by the increase in temperature, so the rate is nearly constant in the 325−340 K temperature range. Eventually the decrease in C_2H_2 becomes so important that the rate falls below that of C_2H_4, so at high conversion values there is a marked decrease in selectivity.

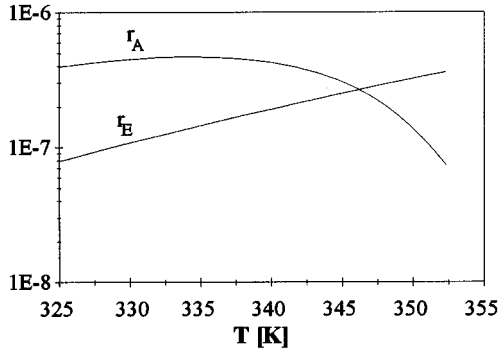


Figure 2. Acetylene and ethylene hydrogenation rate vs temperature for an adiabatic reactor. Kinetic equations of Table 8. Feed mixture as given in Table 3. $T_0 = 325$ K, $P = 16.3$ atm; $C_{CO} = 1400$ ppm.

It should be noted that our kinetic equations do not predict a change in selectivity with the CO content because the r_A/r_E ratio depends only on the concentration of C_2H_2 and C_2H_4. In other words, the presence of CO seems to affect the rate of C_2H_2 and C_2H_4 hydrogenation to the same extent, in accordance with the results of Table 6. As the concentration of CO increases, a higher temperature is required to obtain a given conversion of C_2H_2, but the conversion of C_2H_4 remains the same. The effect of CO may be explained by assuming that it limits the amount of adsorbed H_2 in accordance with kinetic models based on the adsorption of H_2 and CO on the same type of sites. However it is important to stress that the presence of a minimum level of CO is essential to avoid the uncontrolled hydrogenation of C_2H_4.

As mentioned by one reviewer, there is evidence of adsorption of acetylene and ethylene on separate sites which may suggest additional hydrogenation mechanisms. We are aware of several studies that postulated different sites for acetylene and ethylene hydrogenation (All-Ammar and Webb, 1978; McGown et al., 1978; Leviness et al., 1984). Therefore a reaction scheme based on a two-site mechanism could have been included in Table 1. However this model would lead to more complex kinetic equations as shown by Gva and Kuo (1988), increasing the number of adjustable parameters. In addition, our experimental results indicate that the presence of CO prevents the hydrogenation of ethylene up to a very high level of acetylene conversion, as observed in Figure 1. Consequently, a two-site mechanism that may be relevant under different experimental conditions was not used here to obtain additional kinetic equations.

Quite recently Bos et al. (1993b) used rate models from previous studies (Men'shchikov et al., 1975; Gva and Kuo, 1988) to correlate kinetic data obtained with a typical tail-end hydrogenation mixture. The original equations were based on the assumption that the hydrogenation of C_2H_2 and C_2H_4 proceeds on different types of sites, and they were properly modified to take into account the effect of CO. In this case the temperature dependence of the adsorption terms was according to expectations; that is they decrease with an increase

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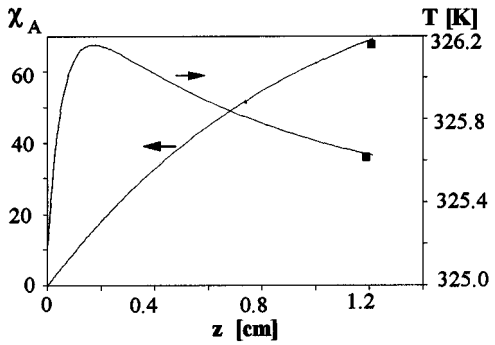


Figure 3. Calculated conversion and temperature profiles for the laboratory reactor using reactor model III and the rate equations of Table 8. Feed mixture as given in Table 3 $T_0 = 325$ K; $P = 16.3$ atm; $C_{CO} = 1400$ ppm. ■, Experimental data.

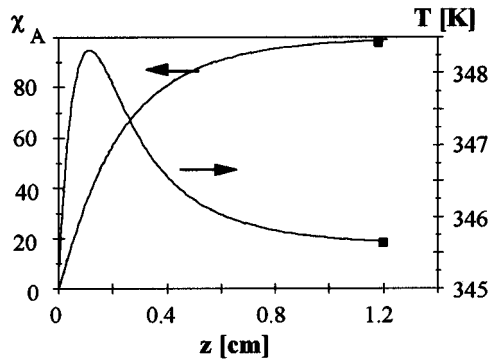


Figure 4. Calculated conversion and temperature profile for the laboratory reactor using reactor model III and the rate equations of Table 8. Feed mixture as given in Table 3, $T_0 = 345$ K; $P = 16.3$ atm.; $C_{CO} = 1400$ ppm. ■, Experimental data.

in temperature. The equations of Bos et al. predict a larger difference between acetylene and ethylene hydrogenation rates due to the reduced partial pressure of hydrogen. They also found that the addition of CO in the 0−60 ppm range does not have the same effect on C_2H_2 and C_2H_4 hydrogenation. The experimental conditions were certainly different from those of the present study, and this situation precludes a better comparison of results. Most of the kinetic studies published to date are related to tail-end hydrogenation processes where low concentrations of CO and H_2 are used.

Finally we need to mention that the absence of pore diffusion limitations in the laboratory experiments was verified using the Weisz−Prater criterion:

$$\Phi = \frac{(-r_A)_{obs}\rho_{cat}d_p^2}{D_{eA}C_{sA}} < 1 \qquad (26)$$

where the observable rate $(-r_A)_{obs}$ was calculated from eq 24 at different conversion levels. Only at $\chi_A > 90\%$ does the Φ module become greater than 1, due mainly to the low value of C_{sA}. For C_2H_4 hydrogenation the criterion is always satisfied because $-r_E < -r_A$ and the conversion is less than 10−15%.

Comparison of Reactor Model III with Experimental Data. Computer simulations of the laboratory reactor, using model III and the kinetic equations of Table 8, are presented in Figures 3 and 4 where the conversion and the axial temperature are plotted as a function of the bed length. The predicted C_2H_2 and C_2H_4 conversions at the reactor end are in good agreement with the experimental results. On the other hand, the temperature profiles show the presence of a hot spot near the reactor entrance. A sharp rise in temperature profiles is followed by a slow decrease along the axial coordinate. The temperature overshoot depends on the feed-reactor wall temperature: at 325 K, with a conversion of 70.7% it is about 1.5 K. Increasing the inlet temperature to 345 K, the C_2H_2 conversion was close to 98% and the temperature overshoot exceeded 3 K. The simulations also indicate that a temperature gradient of about 0.5 K, between the exit gas and the reactor wall, should be expected. As mentioned in the Experimental Section, temperature differences of this magnitude were observed in our runs, which shows the convenience of using a nonisothermal model to fit the experimental data.

As described above, reactor model III takes into account the presence of temperature and concentration differences between the catalyst particle and the gas phase. The calculated concentration gradients were found to be negligible for all runs. On the other hand, the $\Delta T = T_s - T$ values were found to depend on the gas phase temperature which in turn is related to the rate of reaction. At the reactor entrance, in the hot spot region, large temperature gradients between the gas phase and the catalyst particles were calculated. In Figure 3, at 326 K the ΔT value was about 8 K. It increased to nearly 20 K when the gas phase temperature overshot to 348 K, as was the case in Figure 4. These results confirm the suitability of reactor model III to treat the laboratory data in order to obtain the kinetic parameters.

Industrial Reactor Simulation. In order to check the validity of our kinetic equations to predict conversion and temperature profiles, an attempt was made to simulate the operation of an industrial acetylene hydrogenation reactor. The process scheme consists of an ethane cracker followed by three adiabatic reactors in series in a typical front-end hydrogenation configuration. The simulation was restricted to the first unit. The reactor has a diameter of 170 cm and a length of 180 cm, with a catalyst mass of 4290 kg. A typical feed mixture consists of 4.30 wt % H_2, 32.94 wt % C_2H_6, 51.96 wt % C_2H_4, 0.48 wt % C_2H_2, 5.33 wt % CH_4, 0.08 wt % C_3H_4, 1.37 wt % C_3H_6, 1.67 wt % C_4H_6, and 0.04 wt % CO as main components. The flow rate was about 57 900 kg/h, the total pressure was 35 atm, and the inlet temperature was 342.5 K. The pressure drop was quite negligible. These conditions allow us to simulate the reactor with a plug flow, pseudohomogeneous, isobaric and adiabatic model. The high flow velocity eliminates the need for external mass transport considerations, and the absence of internal concentration gradients was already demonstrated using the laboratory reactor data. Consequently the industrial reactor can be described by eq 6 and a simplified form of eq 7:

$$\frac{d(vC_i)}{dz} = (-r_i)\rho_{cat} \qquad (6)$$

$$c_p\frac{dT}{dz} = \rho_{cat}\sum_i(-\Delta H_i)(-r_i) \qquad (7')$$

The simulation results are presented in Figures 5 and 6. The predicted conversion of C_2H_2 is close to 100%, but the measured value in the plant reactor was around 77%. In addition the actual C_2H_4 loss was lower than expected. On the other hand, an exit temperature of 364.5 K was calculated, which is close to the measured value of 363.2 K. The pronounced disagreement be-

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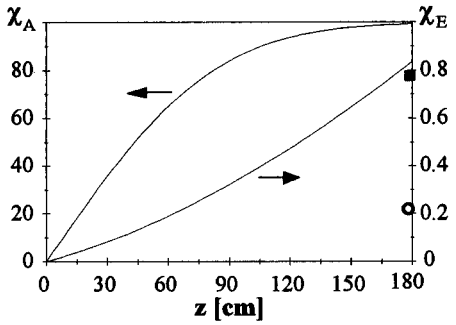


Figure 5. Simulation of an industrial reactor. Acetylene and ethylene conversion axial profiles for a model II adiabatic reactor. ■, C_2H_2 conversion; ○, C_2H_4 conversion.

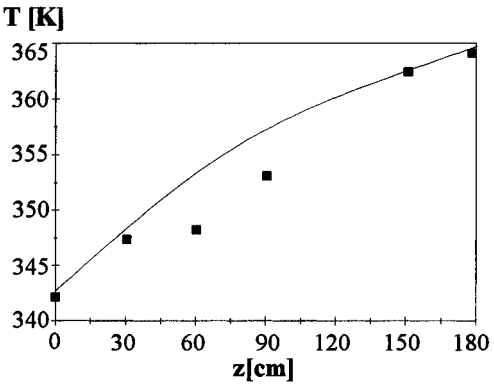


Figure 6. Simulation of an industrial reactor. Computed and measured temperature profiles. ■, Plant data.

tween the estimated and measured conversion values and the approximate temperature profile is due to the large concentration of 1,3-C_4H_6 in the feed mixture. Industrial practice indicates that the hydrogenation of 1,3-C_4H_6 takes place simultaneously with C_2H_2 hydrogenation. The conversion of 1,3-C_4H_6 for the operating conditions mentioned above was around 44%, and this reaction limits the activity for C_2H_2 hydrogenation.

The good agreement between the predicted and measured temperature profiles is due to compensation of the heat of reaction of unconverted C_2H_2 by that produced in the hydrogenation of 1,3-C_4H_6. A certain degree of catalyst deactivation, due to green oil formation, may also have altered the results. A suitable equation for the rate of 1,3-C_4H_6 hydrogenation must be developed in order to predict the observed conversion values in the industrial reactor. As an alternative, we have developed an empirical model for butadiene and propylene + methylacetylene hydrogenation, based on industrial data. By including these equations in the mathematical model of the industrial reactor and also adding a deactivation parameter, a good correspondence between the steady state experimental data and simulation results was obtained (Schbib et al., 1994).

Conclusions

Rate equations for C_2H_2 and C_2H_4 hydrogenation on a commercial Pd/α-Al_2O_3 catalyst have been obtained for reaction conditions similar to those of a front-end hydrogenation process. Experimental data as a function of temperature, pressure, and CO content were obtained in a laboratory reactor covering a wide C_2H_2 conversion range. To our knowledge, kinetic studies of this kind have not been reported in the open literature. The rate equations reflect (i) that the rate of C_2H_2 hydrogenation exceeds that of C_2H_4 hydrogenation up to a high level of conversion and (ii) that the presence of CO is a necessary condition to avoid a runaway situation but the selectivity is not dependent on the CO concentration.

Computer simulation of the laboratory reactor using a nonisothermal model that accounts for interfacial gradients provides a good correlation of experimental data and indicates the existence of a temperature overshoot near the reactor entrance. We were not able to reproduce the conversion profile of an industrial adiabatic converter due to the presence of a large amount of 1,3-C_4H_6 in the cracked gas mixture.

Nomenclature

A = heat transfer area per unit length of the reactor, m^2
A_i = preexponential factor in Arrhenius expression, $mol^{n-1}/(m^{3(n-1)}\ s)$
a_m = specific interfacial surface area
C_i = concentration of ith component in the gas phase, mol/m^3
C_i^* = dimensionless concentration, C_{si}/C_i
C_{si} = surface concentration of component i, mol/m^3
c_p = average heat capacity, J/(mol K)
D_e = effective diffusivity
D_{io} = Damkohler number
d_p = pellet diameter, m
d_t = inside diameter of tube, m
d_{to} = outside diameter of tube, m
E_i = activation energy of the component i, J/mol
F_i = molar flow rate of component i, mol/s
$(-\Delta H)_i$ = heat of reaction, J/mol
h = heat transfer coefficient, $J/(m^2\ K\ s)$
h_w = wall heat transfer coefficient, $J/(m^2\ K\ s)$
h_{wf} = wall/fluid heat transfer coefficient, $J/(m^2\ K\ s)$
h_{fs} = fluid/solid heat transfer coefficient, $J/(m^2\ K\ s)$
k_i = Arrhenius type rate constant of component i
K_i = equilibrium constant of component i
k_{mi} = mass transfer coefficient, m/s
k_p = pellet conductivity, J/(m K s)
k_{ref} = radial effective conductivity, J/(m K s)
k_{rf} = radial conductivity of the fluid, J/(m K s)
k_{rs} = radial conductivity of the solid, J/(m K s)
k_{si} = Arrhenius type rate constant of component i at the surface
K_{Si} = equilibrium constant of component i at the surface
k_w = wall conductivity, J/(m K s)
Pr = Prandt number
Re = Reynolds number
r_i = rate of reaction of component i, $mol/(kg_{cat}\ s)$
[S] = concentration of unoccupied sites
Sc = Schmidt number
[S_t] = concentration of free and occupied sities
SV = space velocity, cm^3 STP $g_{cat}^{-1}\ min^{-1}$
T = temperature in the gas phase, K
T^* = dimensionless temperature of the fluid, T_S/T
T_s = surface temperature, K
T_w = wall temperature
U = overall heat transfer coefficient, $J/(m^2\ K\ s)$
v = average velocity of the fluid through the bed, m/s
w = mass of catalyst, kg
y_i = molar fraction of component i
z = reactor length coordinate, m

Greek Letters

ϵ = void fraction of packed bed
ρ = fluid density, kg/m^3
ρ_{cat} = apparent catalyst density
χ_i = experimental conversion of component i
$\langle\chi_i\rangle$ = calculated conversion of component i
η_i = effectiveness factor of component i

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Subscripts

A = acetylene
CO = carbon monoxide
E = ethylene
Et = ethane
H = hydrogen

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